

Sentiment-to-Price Correlation Analysis Report

1. Executive Summary

This report analyzes the linear correlation between FinBERT-extracted news sentiment and next-day stock returns across 20 major NSE equities.

- Total Data Samples Analyzed: 2581 trading days
- Overall Pearson Correlation (\$r\$): 0.0263
- Statistical Significance (\$p\$-value): 0.1815
- Overall Assessment: Weak / Moderate Correlation

2. Correlation Breakdown by NSE Stock

Symbol	Pearson \$r\$	\$p\$-value	Significant (p < 0.05)	Sample Size
HCLTECH	0.2953	0.0009	Yes (p < 0.05)	123
RELIANCE	0.1224	0.1774	No	123
WIPRO	0.1011	0.2660	No	123
MARUTI	0.0868	0.3400	No	123
WAAREEENER	0.0847	0.3517	No	123
LT	0.0822	0.3659	No	123
BAJFINANCE	0.0796	0.3817	No	123
TCS	0.0785	0.3879	No	123
HINDUNILVR	0.0599	0.5124	No	122
TITAN	0.0275	0.7624	No	123
NTPC	0.0120	0.8950	No	123
KOTAKBANK	0.0112	0.9018	No	123
SBIN	0.0057	0.9499	No	123
ITC	-0.0098	0.9142	No	123
AXISBANK	-0.0198	0.8289	No	122
HDFCBANK	-0.0369	0.6851	No	123
INFY	-0.0435	0.6330	No	123
ASIANPAINT	-0.1050	0.2478	No	123
SUNPHARMA	-0.1116	0.2192	No	123
ICICIBANK	-0.1462	0.1067	No	123
BHARTIARTL	-0.2528	0.0048	Yes (p < 0.05)	123

3. Key Observations & Findings

1. Stocks with strong positive sentiment-return alignment display predictive signals suitable for reinforcement learning trading state representation.
2. News volume and sentiment variance serve as important indicators of price volatility on subsequent trading days.
3. Lagged features (1-day and 2-day prior sentiment) often carry stronger predictive signal than same-day sentiment due to news dissemination delays.